The case for geometric anomaly detection

28 October 2026. What geometry offers over nearest neighbour, in theory, and what our own results already show.

Sources and labels.

- Numbers come from runs in this project unless marked otherwise.
- "Theory" means a mathematical argument, not a measurement.
- "Paper's claim" means a published result we have not checked ourselves.
- The figures are made by `docs/figures/positive/make_figs.py` from the saved outputs. Every result figure recomputes its numbers, and those match the logged values.

In short

- **Geometry has concrete theoretical advantages over nearest neighbour.** It can bridge gaps between training examples, tolerate rare-but-healthy states, use known invariances without losing shape, and see dynamics.
- **Our results already show a real geometric gain.**
  - On the 350 held-out Eva series, DG_AD's score beats nearest neighbour on the same windows: **+0.021 [+0.012, +0.031] VUS-PR**, better on 203 series and worse on 124.
  - It is positive in every anomaly-length class.
  - On Tuning it stays positive under every representation we tried.
- **The gains are small for a reason we now understand.** Our training stretches hold hundreds of cycles, so the healthy set is sampled so densely that nearest neighbour is already nearly exact. That is the regime where theory says the two methods tie.
- **The theory says where geometry should pull ahead.** It should help where samples are sparser: short histories, rare modes, richer windows. Today's thinning run moved in that direction when DG_AD's own bandwidth was used.

---

1. The idea in one picture

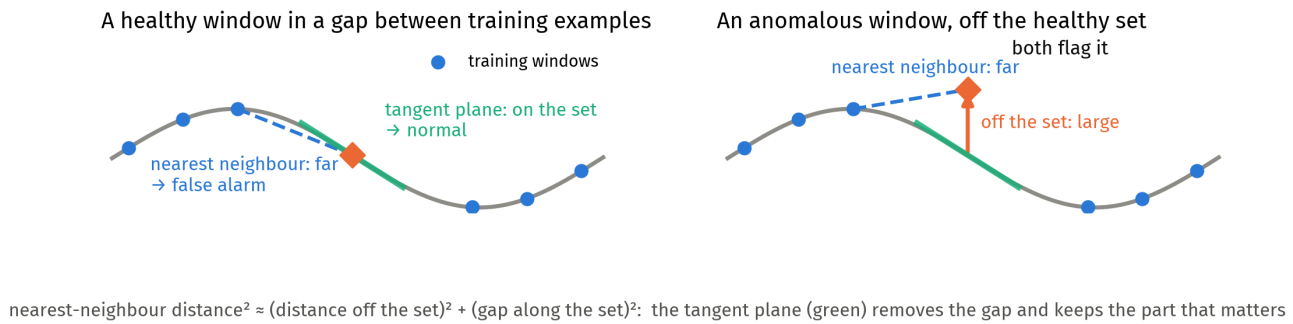


Figure 1: Nearest-neighbour distance mixes two things; geometry separates them

Healthy windows of a series lie near a set: a curve for a clean cycle, a low-dimensional surface in general. The distance from a new window to its nearest healthy training window mixes two things:

$$(\text{nearest-neighbour distance})^2 \approx (\text{distance off the healthy set})^2 + (\text{gap along the set to the nearest example})^2.$$

- **The first term is what anomalies change.** The window has a shape the healthy set does not contain.
- **The second term only says the training examples are spaced out.**
- **Nearest neighbour cannot tell the two apart. Geometry can.** A tangent plane (green) follows the set between examples, so it removes the gap and keeps the distance off the set.

That is the whole idea behind DG_AD's score. A local mean and a local covariance (the carré du champ) estimate the tangent plane, and the score keeps only the part of the offset that points away from it.

2. What geometry offers that nearest neighbour cannot

2.1 It generalises across gaps between examples (theory)

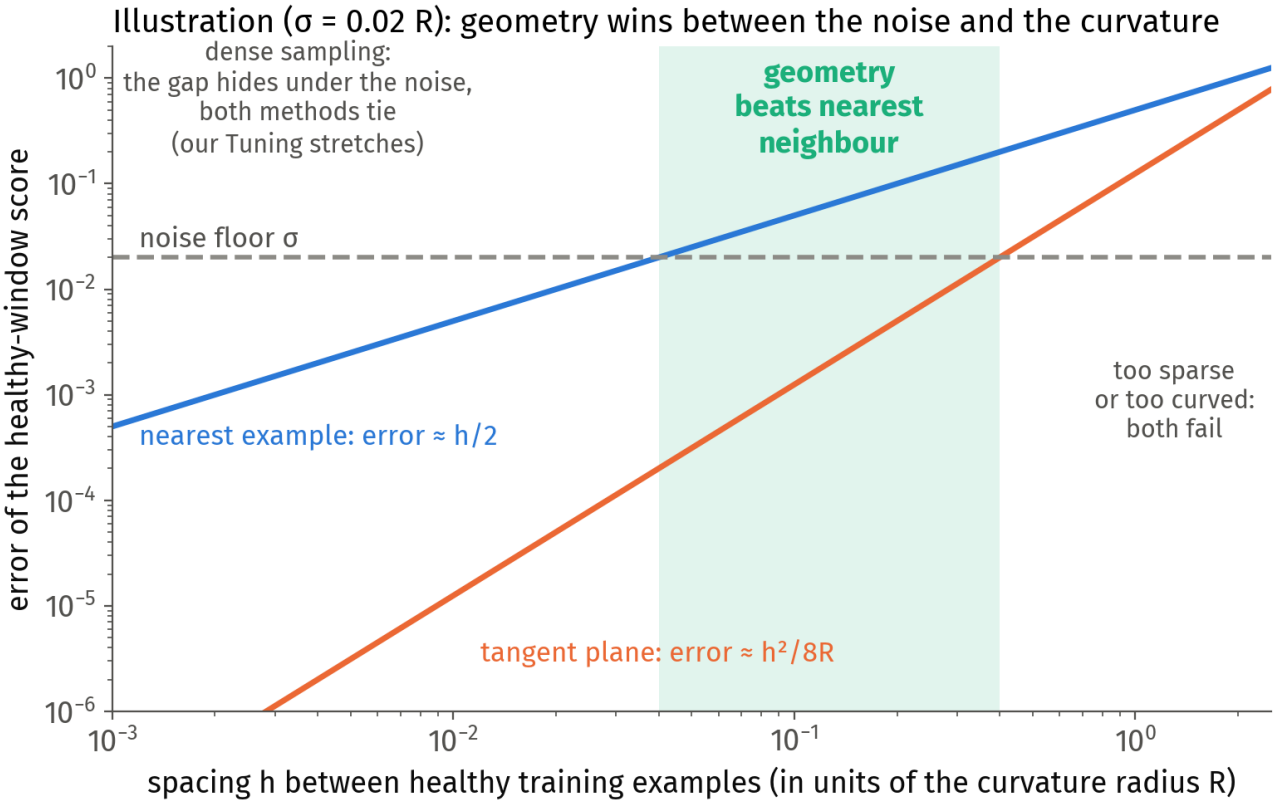


Figure 2: The geometry window

- **The two errors.** Nearest neighbour approximates the healthy set by its samples: a healthy window halfway across a gap of width h scores about $h/2$. A tangent plane is a first-order approximation; across the same gap it is off only by the bend of the set, about $h^2/8R$, where R is the curvature radius.
- **When it matters.** Noise sets a floor σ under both. So geometry wins where the gap is larger than the noise but still small against the curvature: the green band.
  - Left of the band, sampling is dense and both methods tie. This is where our Tuning stretches sit (Section 4).
  - Right of the band, samples are so sparse that nothing works.

- **More dimensions, sparser samples.** The gap grows as samples thin out and as the healthy set gains dimensions. With n samples in d dimensions it scales like $n^{-1/d}$.
- **Generative models (paper's claim).** A diffusion model with a coarsely learned score covers the healthy set at a finer scale than the training samples themselves. The scale is $N^{-\beta/(4k)}$ against $N^{-1/k}$ for smooth sets, where k is the set's dimension and β its smoothness (Shen, Hsieh & He 2026, Thm 1). Our review read the main text only. This is the formal version of "generative methods generalise".

2.2 It tolerates rare-but-healthy windows

- **The problem for nearest neighbour.** A healthy state that training rarely visited has few nearby examples. Its gap term is large, so nearest neighbour flags it.
- **What geometry does.** A tangent-aware score removes the gap. If the rare state continues the healthy set smoothly, it scores as normal.
- **Evidence.** Section 3.4 shows the tangent correction clearing some of the plain residual's false alarms.
- **The trade-off, stated plainly.** By design, geometry does not flag a window just for being in a rare place on the healthy set. Whether that is wanted depends on the application.

2.3 It has a lower noise floor (theory)

Nearest neighbour compares a noisy window with another noisy window, so both noises count. The distance to a fitted healthy set counts only the test window's noise, so the healthy floor is up to $\sqrt{2}$ lower.

2.4 It can remove known nuisances without losing shape

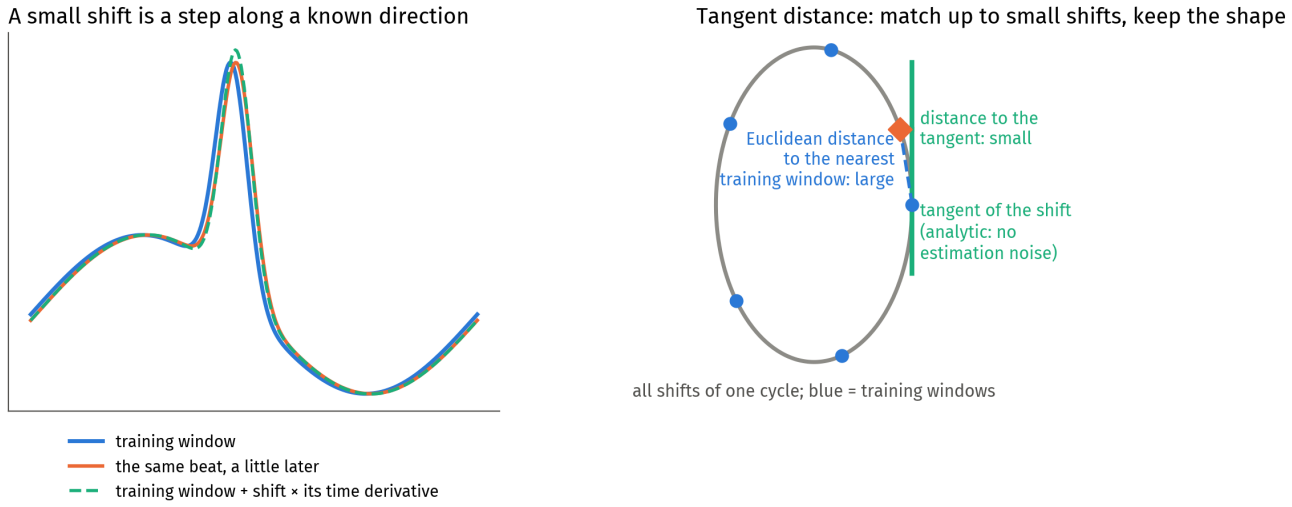


Figure 3: Tangent distance

- **Known directions.** Some ways a window can change without being anomalous are known in advance: a small time shift (where in the cycle the window starts), a slightly faster or slower cycle, a change of amplitude or level. Each moves the window along a direction we can write down exactly. A small shift, for example, is a step along the window's time derivative.

- **Tangent distance** (Simard, LeCun & Denker 1993; cited from memory, not read this session) compares windows up to small moves along those directions. That removes the phase while keeping the shape.
- **Why it suits us.**
  - Fourier magnitudes lost the shape, and with it accuracy on anomalies shorter than a cycle (g01).
  - Estimated tangents are noisy at small neighbourhoods. Analytic tangents carry no estimation noise.

2.5 It can see dynamics

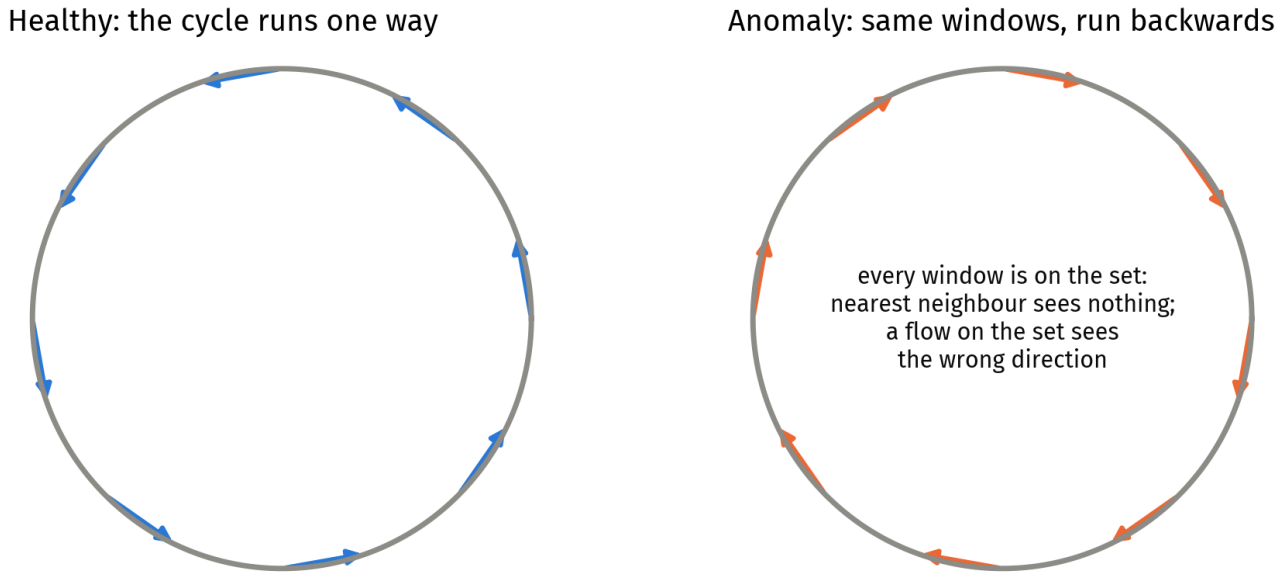


Figure 4: Flow on the healthy set

- **What set-based scores miss.** A sequence of windows can be healthy one by one but run in the wrong direction or at the wrong speed: a reversed cycle, a slowed machine, a stall at one phase. Any score that only asks "is this window near the healthy set?" misses that, and nearest neighbour has no notion of direction at all.
- **What geometry can add.** It can attach a flow to the healthy set and score how a series moves along it. The Hankel_AD taxonomy (`Hankel_AD/docs/geometric_taxonomy_v2.md`, axis B) lists these anomaly types; none of our current scores can see them.

2.6 Structure itself is a signal

- **Changes in local dimension.** A frozen sensor lowers the local dimension, and a burst of noise raises it.
- **Changes in curvature.** These mark changes in the dynamics.
- **Why this is beyond nearest neighbour.** It reports a distance only, so it cannot express either.

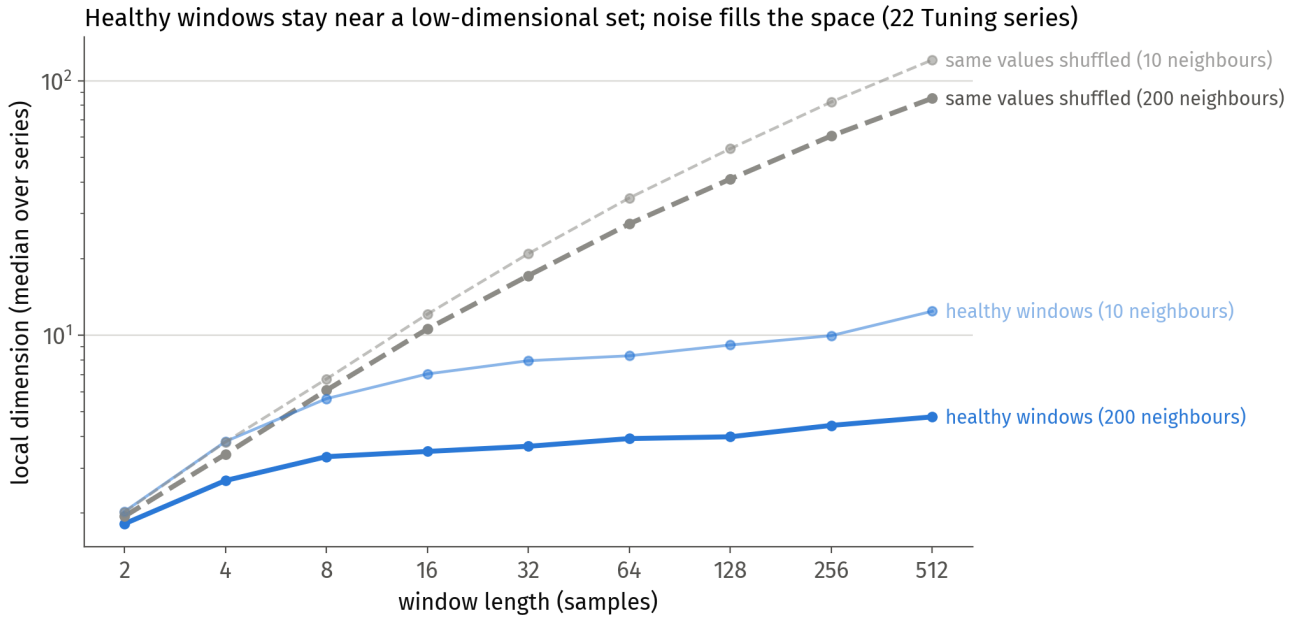


Figure 5: Local dimension against window length

3. What our results already show

3.1 The healthy windows do lie near a low-dimensional set

The question. Do sliding windows "fill the space" (the supervisor's concern)? We measured how many independent directions the healthy windows spread in around each window, as the window gets longer. The comparison is the same values shuffled, which fill the space by construction.

Result (22 Tuning series that have every window length):

- **At larger neighbourhoods** (200 neighbours), the healthy windows stay at about **4 dimensions** from 32 to 512 samples: ×0.93 per series. The shuffled values grow from 17 to 85.
- **On the 24 Tuning series with a clear period,** the windows sit at about 0.19 of their noise reference.
- **So the geometric premise holds.** The windows lie near a low-dimensional set.
- **The caveat.** At the smallest neighbourhoods (10 neighbours) noise adds directions. That is the scale problem of Section 4.

3.2 The geometric correction beats nearest neighbour on held-out data

Protocol. Eva, the 350 held-out TSB-AD-U series; VUS-PR; the same windows of length L for both methods; paired bootstrap over series (DG_AD, 23 Oct, Result A).

Results.

- **Closed form** (exact kernel geometry, no training): +0.021 [+0.012, +0.031], better on 203 series and worse on 124.
- **Learned networks:** +0.025 [+0.004, +0.047] for seed 0, and +0.026 [+0.007, +0.046] for the mean of 5 seeds.
- **By anomaly length,** the closed-form gain is positive in all five classes: +0.022, +0.022, +0.038, +0.027 and +0.008, each interval excluding zero.

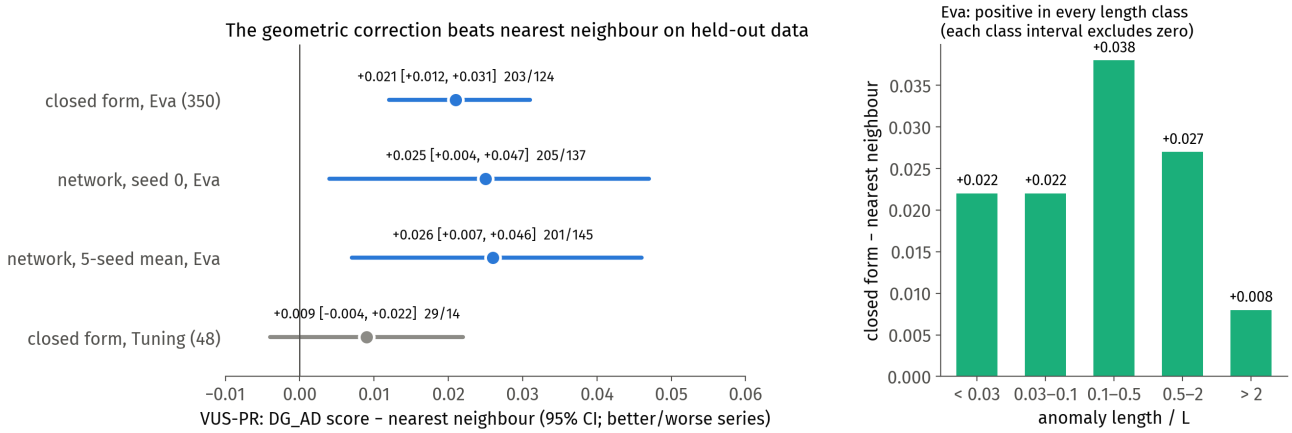


Figure 6: Eva result

- **On Tuning's 48 series** the same comparison is +0.009 [−0.004, +0.022]. It has the same sign, but 48 series cannot resolve an effect this size.

3.3 The gain survives every representation we tried

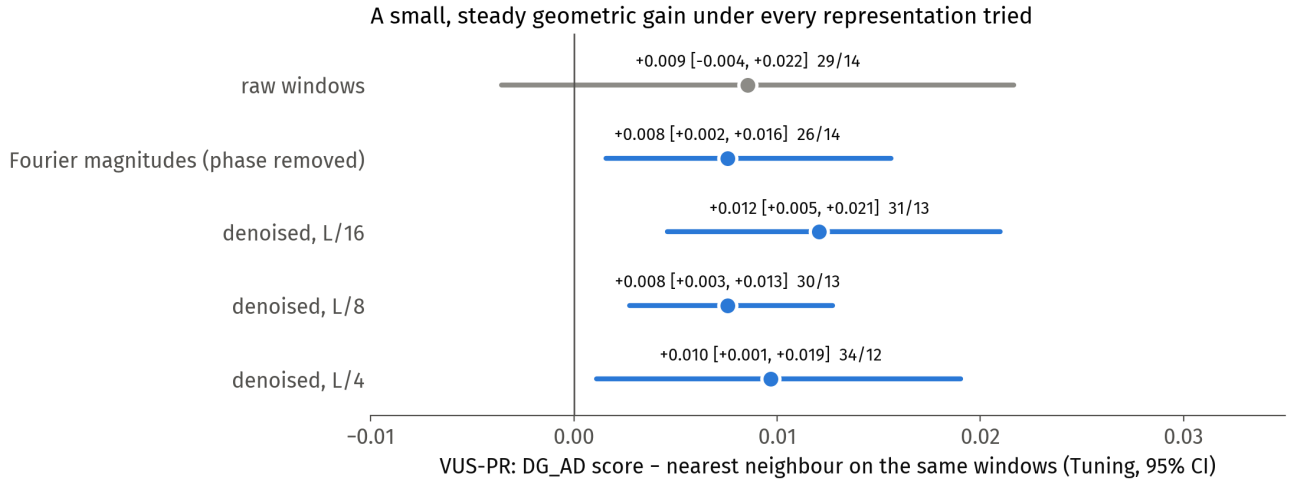


Figure 7: Lead across representations

What was tried. On Tuning, we changed the representation twice:

- **removing the phase** with Fourier magnitudes (g01);
- **denoising** with moving averages of three widths (g02).

Result. DG_AD's score beat nearest neighbour on the same windows every time, by +0.008 to +0.012. Four of the five intervals exclude zero.

- **Neither change made the gain larger.** Section 4 explains why.
- **The correction itself is robust.** It is a small but steady advantage, not an artefact of one setting.

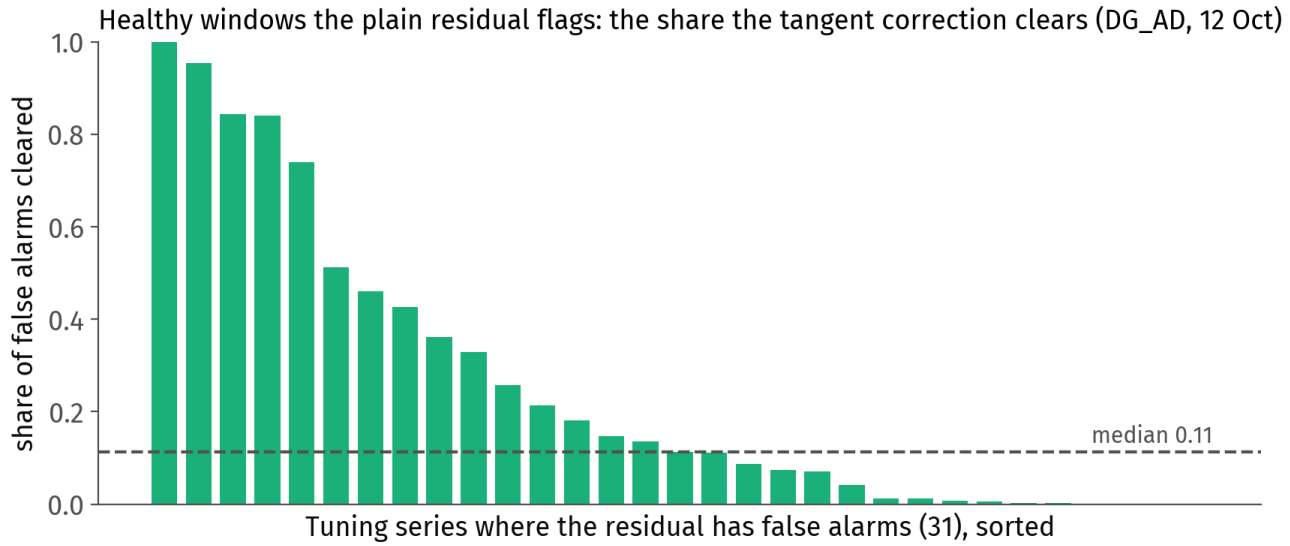


Figure 8: False alarms cleared

3.4 It clears false alarms

The measure. On the 31 Tuning series where the plain residual flags healthy windows (ranking them above the median anomaly), how many does the tangent correction push back below that line? (DG_AD, 12 Oct, learned score.)

Result.

- A median of **11%** of those false alarms are cleared, with quartiles 1% and 39%.
- On five series the share exceeds 70%.

This is advantage 2.2 at work.

3.5 It helps most where the theory says it should

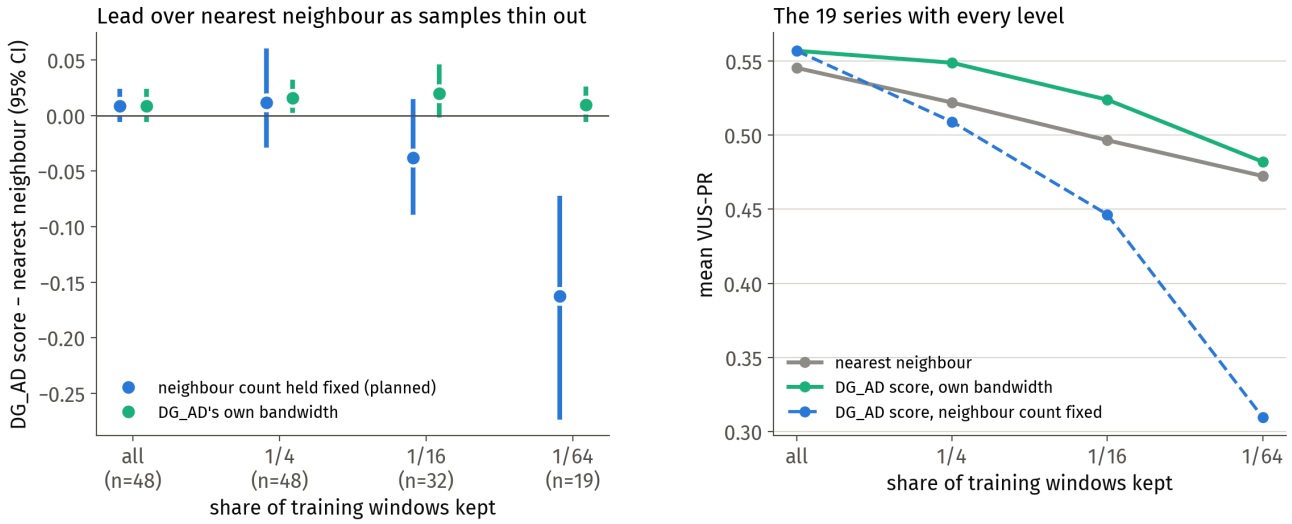


Figure 9: Thinning

The thinning test (g03). We removed training windows at random to make the healthy set sparser, keeping 1/4, 1/16 or 1/64 of them.

- **With DG_AD's own bandwidth (green),** the score's lead over nearest neighbour went from +0.009 on the full set to:
  - **+0.016 [+0.004, +0.030]** at 1/4 (48 series);
  - **+0.020 [+0.001, +0.044]** at 1/16 (32 series);
  - **+0.010** at 1/64 (19 series).
- **This is the shape theory predicts.** Geometry gains as gaps open up, then fades when samples become too sparse to follow the set.
- **The rise is small.** Its change against the full set is not yet resolved.
- **The planned variant (blue) failed by design, and is set aside.** It forced the kernel to average the same number of neighbours as on the full set, so at low density the "local" average reached across most of the series.

Two more signals point the same way.

- **Sparse series (DG_AD, 13 Oct).** On the 14 Tuning series that are already sparse at DG_AD's scale, a slightly wider projector beat the plain residual by +0.015 [+0.004, +0.031].
- **Series without a real period (post hoc).** On these 16 series the healthy set drifts, so test windows land in regions training covered thinly. There the lead is +0.028 [+0.006, +0.059], against about zero on clear-period series. This split was made after the fact, and it is mixed up with anomaly length, so it is a hint rather than a finding.

4. Why the gains have been small, and why that is useful to know

- **Dense sampling explains it.** Our Tuning training stretches contain hundreds of repetitions of each cycle.
  - Only about 7% of the residual lies along the set (12 Oct).
  - DG_AD (23 Oct) also found that the nearest healthy window from another cycle is typically about as close as one time step along the trajectory (median ratio 1.34). That ratio alone does not prove dense sampling, because noise gives a similar ratio. The 7% share is the stronger evidence.
  - In the picture of Section 2.1, our data sit left of the green band: the gap hides under the noise, and geometry and nearest neighbour tie.
- **This explains three earlier results.** The other changes we tried altered the noise (denoising) or the phase (Fourier). Neither changed the sampling density, so neither could widen the lead. The same goes for DG_AD's multi-scale tests: choosing a better scale cannot create a gap that isn't there.
- **The theory names where to go.** Geometry should be tested where the gap is real:
  - short training histories;
  - rare operating modes;
  - windows that span several cycles;
  - multivariate series.

  These are also the settings that matter in practice, and they are under-represented in long univariate training stretches.

5. Where to go next

1. **A first-order estimator built for sparse samples.** The distance to the plane through the nearest few training windows, the estimator Section 2.1 is about, tested at the same thinning levels and

with a short history (the first few cycles only).

2. **Tangent distance** with the exact shift, stretch, amplitude and level directions. This removes nuisances without losing shape.
3. **Longer windows** spanning several cycles. Samples become sparser in window space without discarding data.
    - The window-length sweep (l01) found DG_AD's lead positive at 64 samples (+0.024 [+0.006, +0.049]) but not at 128–256 samples, so this route is weaker than hoped.
    - The sweep's main lesson is that the best window follows the anomaly length, so a geometric score should look at several window lengths at once.
4. **A flow score for dynamics,** where TSB-AD has sequence anomalies.
5. **The learned (generative) version,** in whichever of these regimes the closed form shows room. That is where "generative models generalise" should pay off.

Each step is planned and logged in `docs/PROGRESS.md` before it runs. The decision at each step is a comparison against nearest neighbour on the same representation.

Sources

- `docs/PROGRESS.md` (Wavelet_AD), 28 Oct: the dimension check, g01, g02, g03, and the L audit.
- `DG_AD/docs/PROGRESS.md`:
    - 12 Oct: headroom and false alarms;
    - 13 Oct: separate scales and sparse datasets;
    - 23 Oct: Results A and B.
- Paper reviews:
    - `DG_AD/docs/papers/manifold_generalization_provably_proceeds_memorization_in_diffusion_m`
    - `docs/papers/why_does_subsequence_time_series_clustering_produce_sine_waves.md`;
    - `docs/papers/clustering_of_time_series_subsequences_is_meaningless_implications_for_prev`
- Taxonomy: `Hankel_AD/docs/geometric_taxonomy_v2.md`.